

Computability & Logic

June 2023

For questions that allow multiple answers, you have to select ALL correct answers in order to get maximal points.

Which of the following rules can appear in a context-free grammar in Chomsky Normal Form?

(alphabet $\Sigma = \{0, 1\}$, non-terminals $\Gamma = \{S, A, B\}$)

Choose one answer

☐ $S \rightarrow A S B$

☐ $S \rightarrow A$

☐ $S \rightarrow A B$

☐ $S \rightarrow 01$

Below, we consider the following decision problems:

- **Accepts**: given input TM T and word v , decide if T accepts v
- **Halting**: given input TM T and word v , decide if T halts on v
- **Accepts- Λ** : given input TM T , decide if T accepts the empty word
- **Halts-on- Λ** : given input TM T , decide if T halts on the empty word

Now given a Turing Machine M and a word w , we define the computable reduction function $F(M, w)$ that returns the TM that behaves as " $write(w) \rightarrow PB \rightarrow G(M)$ ", that is, $F(M, w)$ returns a TM that writes w on the tape, goes back to the previous blank symbol, and runs $G(M)$ on w . Here $G(M)$ replaces the reject-state h_r in M by a new state with only self-loops.

Which problem reduction is achieved by F ?

Choose one answer

- ☐ F reduces "Halts-on- Λ " to "Accepts"
- ☐ F reduces "Halting" to "Accepts- Λ "
- ☐ F reduces "Accepts" to "Halts-on- Λ "
- ☐ F reduces "Accepts- Λ " to "Halting"

Which of the following statements are correct?
(equivalent = accepts the same language)

Choose one or more answers

- ☐ Every non-deterministic *Turing Machine* can be transformed to an equivalent deterministic one.
- ☐ Every non-deterministic *Finite Automaton* can be transformed to an equivalent deterministic one.
- ☐ Every ambiguous *Context-Free Grammar* can be transformed to an equivalent non-ambiguous one.
- ☐ Every non-deterministic *Push-Down Automaton* can be transformed to an equivalent deterministic one.

Consider the following proof in natural deduction for propositional logic:

1.	$\neg(p \wedge q)$
2.	p
3.	q
4.	$p \wedge q$
5.	$\perp$
6.	r
7.	$q \rightarrow r$
8.	$p \rightarrow q \rightarrow r$
9.	$\neg(p \wedge q) \rightarrow (p \rightarrow q \rightarrow r)$

Which of the following proof rules are used in this proof? (select all correct answers)

Choose one or more answers

- ☐ $\wedge$ -introduction
- ☐ $\wedge$ -elimination
- ☐ $\rightarrow$ -elimination
- ☐ $\rightarrow$ -introduction
- ☐ $\neg$ -elimination
- ☐ proof by contradiction
- ☐ $\neg$ -introduction

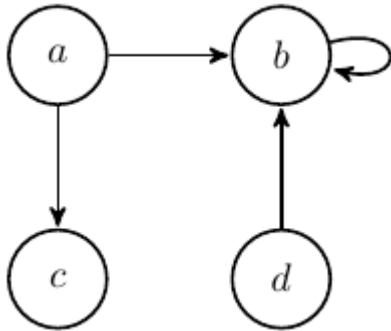
Which of the following Hoare triples is valid?

Choose one answer

- ☐ $\{x + y \leq 4\} \ z := x + y; x := 3 * z \ \{x \leq 10\}$
- ☐ $\{x + y \leq 3\} \ z := x + y; x := 3 * z \ \{x \leq 10\}$
- ☐ $\{x + y \leq 3\} \ x := 3 * z; z := x + y \ \{x \leq 10\}$
- ☐ $\{x + y \leq 4\} \ x := 3 * z; z := x + y \ \{x \leq 10\}$

We consider predicate logic with a binary relation symbol T . We look at a particular model with universe $A = \{a, b, c, d\}$.

The relation is interpreted as in the following diagram, where $T(x, y)$ holds if x has an arrow to y .



Which of the following predicate-logic formulas does NOT hold in this model?

Choose one answer

- ☐ $\exists x \forall y \neg T(x, y)$
- ☐ $\forall x \exists y \neg T(x, y)$
- ☐ $\forall x \forall y \forall z (T(x, y) \wedge T(x, z) \rightarrow y = z)$
- ☐ $\forall x \forall y \forall z (T(x, y) \wedge T(y, z) \rightarrow T(x, z))$

Which of the following formulas in predicate logic are a tautology?

Choose one or more answers

☐ $(\forall x P(x) \wedge \forall x Q(x)) \rightarrow \forall x (P(x) \wedge Q(x))$

☐ $\forall x (P(x) \wedge Q(x)) \rightarrow (\forall x P(x) \wedge \forall x Q(x))$

☐ $\exists x (P(x) \wedge Q(x)) \rightarrow (\exists x P(x) \wedge \exists x Q(x))$

☐ $(\exists x P(x) \wedge \exists x Q(x)) \rightarrow \exists x (P(x) \wedge Q(x))$

Which statement on non-ambiguous grammars is *wrong*?

Choose one answer

- ☐ Every word has one derivation tree
- ☐ Every word has one left-most derivation
- ☐ Every word has one derivation
- ☐ Every word has one right-most derivation

The following problems all take a Turing Machine M as input. To which problem can we apply Rice's theorem to conclude undecidability?

Choose one answer

- ☐ All words in $\mathcal{L}(M)$ have length $|w| > 0$.
- ☐ All words in $\mathcal{L}(M)$ have length $|w| \geq 0$.
- ☐ M never runs off the tape on the left.
- ☐ The number of states in M is even.

Given is a context-free grammar G that is *ambiguous*.

Which one of the following sentences can be concluded?

Choose one answer

- ☐ Every word in the language of G has more than one derivation.
- ☐ There exists a word that has more than one left-most derivation in G .
- ☐ Checking if a word is in the language of G is undecidable.
- ☐ From some non-terminal symbol, G generates more than one word.

Consider the following dominoes for Post's Correspondence Problem:

$$x_1 = \begin{pmatrix} 00 \\ 0 \end{pmatrix}, x_2 = \begin{pmatrix} 00 \\ 0010 \end{pmatrix}, x_3 = \begin{pmatrix} 1 \\ 00 \end{pmatrix}$$

Choose one answer

- ☐ This instance of PCP has solution x_2, x_3, x_1, x_1
- ☐ This instance of PCP has solution x_2, x_3, x_1
- ☐ This instance of PCP has solution x_2, x_3, x_1, x_1, x_1
- ☐ This instance of PCP has no solution at all

Consider an arbitrary regular language R and an arbitrary context-free language C . Which of the following statement(s) can be concluded?
(X' denotes the complement of X).

Choose one or more answers

- ☐ $C' \cap R$ is regular
- ☐ $C' \cap R$ is context-free
- ☐ $C \cap R'$ is regular
- ☐ $C \cap R'$ is context-free

Three of the following regular expressions are equivalent, but one of them generates a different language. Which one is different?

Choose one answer

- ☐ $bb(ab)^*$
- ☐ $b(ba)^*$
- ☐ $b + bb(ab)^*a$
- ☐ $b + bba(ba)^*$

Consider the Hoare triple $\{P\} S \{Q\}$ under partial and total correctness.
Which statement on this triple is correct?

Choose one answer

- ☐ Under total correctness, program S will terminate for states that satisfy precondition P
- ☐ Under partial correctness, program S will terminate for states that satisfy precondition P
- ☐ Under total correctness, program S will always terminate, no matter in which state it starts
- ☐ Under total correctness, program S may not terminate, even for states that satisfy precondition P

Below, SAT is the satisfiability problem for propositional logic, and X is an arbitrary problem in P.

Which of the following statements on polynomial-time reduction are correct?

Choose one or more answers

- ☐ If $SAT \leq_P X$, then X is NP-complete.
- ☐ If $X \leq_P SAT$, then X is NP-complete.
- ☐ If $SAT \leq_P X$, then $P = NP$.
- ☐ If $X \leq_P SAT$, then $P = NP$.

Which of the following statements on the relation between a language and its complement are true?

Choose one or more answers

- ☐ If a language L is context-free, then its complement L' is context-free
- ☐ If a language L is recursively enumerable, then its complement L' is recursively enumerable
- ☐ If a language L is context-free, then its complement L' is recursive
- ☐ If a language L is regular, then its complement L' is context-free

Which of the following formulas in propositional logic are satisfiable?

Choose one or more answers

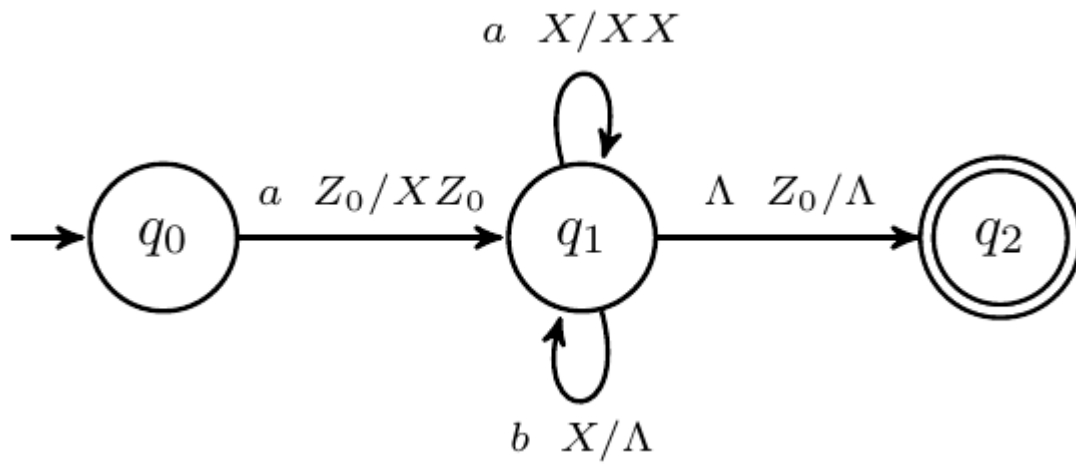
☐ $(p \vee q) \wedge (\neg p \wedge \neg q)$

☐ $(p \wedge q) \vee (\neg p \wedge \neg q)$

☐ $(p \wedge q) \wedge (\neg p \wedge \neg q)$

☐ $(p \vee q) \vee (\neg p \vee \neg q)$

Consider the following PDA:



Which of the following words are accepted (by final state) by this PDA?

Choose one or more answers

☐ $abab$

☐ $aaab$

☐ $abba$

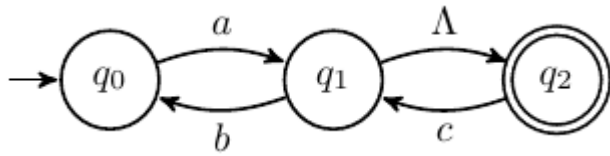
☐ $aabb$

Which statement on the decidability of logic(s) is correct?

Choose one answer

- ☐ *Provability* is decidable for BOTH propositional logic and predicate logic.
- ☐ *Validity* is decidable for BOTH propositional logic and predicate logic.
- ☐ *Provability* is decidable for propositional logic, but NOT for predicate logic.
- ☐ *Validity* is decidable for the logic of arithmetic, but *provability* is NOT.

Consider the following Nondeterministic Finite Automaton M:



Which of the following regular expressions produce exactly the same language as M accepts?

Choose one or more answers

☐ $(ac^*b)^*ac^*$

☐ $a(ba + c)^*$

☐ $ac^*(ba)^*$

☐ $(ab)^*ac^*$

Which of the following problems on Context Free Grammars is undecidable?
(below, w is a word and G_1, G_2 are context-free grammars)

Choose one answer

- ☐ Given G_1, G_2 , is $\mathcal{L}(G_1) \cap \mathcal{L}(G_2) = \emptyset$?
- ☐ Given G_1, G_2, w , is $w \in \mathcal{L}(G_1) \cup \mathcal{L}(G_2)$?
- ☐ Given G_1, G_2 , is $\mathcal{L}(G_1) \cup \mathcal{L}(G_2) = \emptyset$?
- ☐ Given G_1, G_2, w , is $w \in \mathcal{L}(G_1) \cap \mathcal{L}(G_2)$?

Which one of the following languages is *not* Context-Free?
(the alphabet is {a,b,c}).

Choose one answer

- ☐ The set of words that have an equal number of a's, b's and c's.
- ☐ The set of words with all a's before the b's *and* all b's before the c's.
- ☐ The set of all words that start with an "a", have a "b" in the middle, *and* end with a "c".
- ☐ The set of words that have more a's than b's *or* more b's than c's.

Consider the following Context-Free Grammar in Chomsky Normal Form:

$$S \rightarrow AA \mid c$$

$$A \rightarrow SS \mid b$$

We use the CYK-algorithm to parse the word $bbcb$ in this grammar. A partially filled table is already shown.

How much entries will contain A or S , respectively?

A	?	?	?
	A	?	?
		S	?
			A

Choose one answer

- ☐ 4 times A and 3 times S
- ☐ 5 times A and 3 times S
- ☐ 3 times A and 2 times S
- ☐ 5 times A and 5 times S

Consider the following truth tables, for propositional formulas A, B, C and D:

p	q	r	A	B	C	D
T	T	T	F	F	F	F
T	T	F	T	F	F	F
T	F	T	F	F	T	T
T	F	F	F	F	T	F
F	T	T	T	F	F	F
F	T	F	T	T	F	F
F	F	T	F	F	T	F
F	F	F	F	F	F	F

What is the correct truth table for the formula $\neg(q \rightarrow (p \vee r))$?

Choose one answer

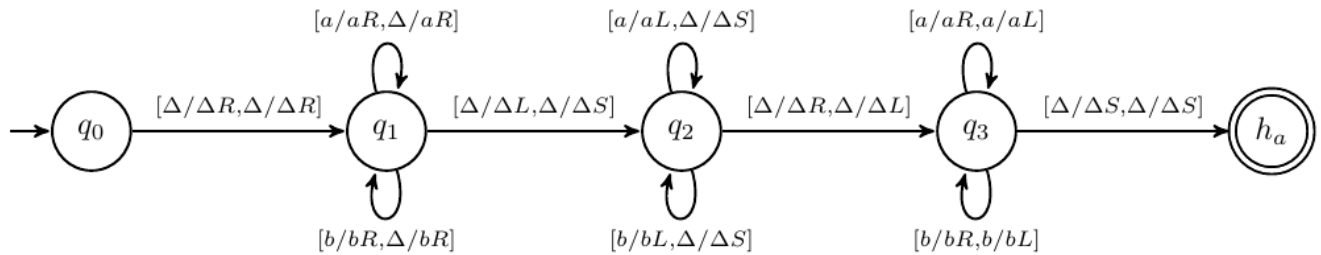
- ☐ D
- ☐ C
- ☐ B
- ☐ A

Given is a vocabulary with function symbols $\{f, g\}$ and predicate symbols $\{A, B\}$. Which of the following is a valid application of the $(\forall e)$ -rule in natural deduction?

Choose one answer

- ☐
$$\frac{\forall x \exists y (A(x, y) \rightarrow B(x))}{\exists y (A(f(y), y) \rightarrow B(f(y)))} (\forall e)$$
- ☐
$$\frac{\forall x \exists y (A(x, y) \rightarrow B(x))}{\exists y (A(f(x), y) \rightarrow B(f(x)))} (\forall e)$$
- ☐
$$\frac{\forall x \exists y (A(x, y) \rightarrow B(x))}{\exists y (A(B(x), y) \rightarrow B(B(x)))} (\forall e)$$
- ☐
$$\frac{\forall x \exists y (A(x, y) \rightarrow B(x))}{\exists y (A(f(x), y) \rightarrow B(g(x)))} (\forall e)$$

What does the following multi-tape Turing Machine achieve?

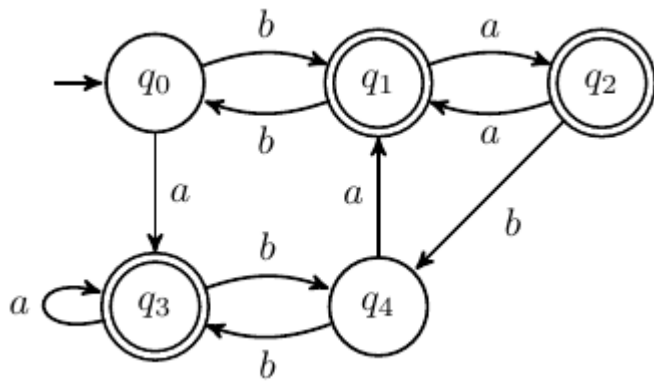


(below, w^R denotes the reverse of word w)

Choose one answer

- ☐ It *decides* the language $\{w w \mid w \in \{a, b\}^*\}$.
- ☐ It *decides* the language $\{w w^R \mid w \in \{a, b\}^*\}$.
- ☐ It *accepts* the language $\{w w^R \mid w \in \{a, b\}^*\}$, but it doesn't *decide* it.
- ☐ It *accepts* the language $\{w w \mid w \in \{a, b\}^*\}$, but it doesn't *decide* it.

Consider the following DFA:

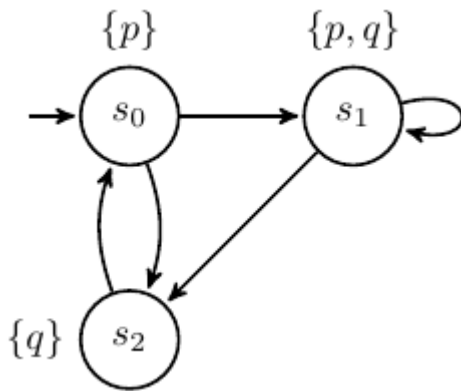


What is the number of states of the corresponding minimal complete DFA?

Choose one answer

- ☐ 3 states
- ☐ 5 states
- ☐ 2 states
- ☐ 4 states

Consider the following Kripke structure:



One of the following LTL properties does *not* hold on this model. Which one?

Choose one answer

- ☐ $p \text{ W } q$
- ☐ $\text{GF } q$
- ☐ $p \text{ U } q$
- ☐ $\text{FG } p$

The following properties each define some language over the alphabet $\{a,b,c\}$. Which of these languages are regular?

Choose one or more answers

- ☐ The total number of a's and b's is even, but the total number of b's and c's is odd.
- ☐ The number of c's exceeds the total number of a's and b's
- ☐ The symbols b and c do not occur next to each other.
- ☐ The length of the word is a square number.

Let e be a reasonable encoding for Turing Machines into words over the alphabet $\Sigma = \{0, 1\}$.

Which statement is true?

Choose one answer

- ☐ The language $\{e(M) \mid M \text{ accepts } 0110\}$ is regular (but not finite)
- ☐ The language $\{e(M) \mid M \text{ accepts } 0110\}$ is context-free (but not regular)
- ☐ The language $\{e(M) \mid M \text{ accepts } 0110\}$ is recursive (but not context-free)
- ☐ The language $\{e(M) \mid M \text{ accepts } 0110\}$ is recursively enumerable (but not recursive)

